ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

CIVIL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Civil Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept other relevant, correct answers. Penalise once per question for missing units.

QUESTION 1

1 1.1 B ✓✓ 1.2 B ✓✓ 1.3 A ✓✓ 1.4 B ✓✓ 1.5 B ✓✓ [10]

QUESTION 2

2.1 Any THREE: provide a safe workplace ✓; provide protective equipment ✓; train workers in safe work methods ✓; maintain machines and equipment; report and investigate accidents; provide first-aid facilities (3)

2.2 (a) safety goggles / face shield (and ear protection) ✓ (b) dust mask and gloves ✓ (c) safety harness ✓ (3)

2.3 Red: prohibition (do not) / fire equipment ✓; blue: mandatory (must do) ✓; yellow: warning / caution ✓; green: safe condition / first aid / emergency exit ✓ (4)

2.4 Wear gloves; press a clean cloth or dressing firmly on the wound to stop the bleeding ✓; raise the hand above the heart ✓; bandage it and get medical help (call for help if the bleeding does not stop) ✓. (3)

2.5 To remove dust, fumes and harmful gases ✓ so that workers can breathe clean air and to reduce the risk of fire ✓. (2)

[15]

QUESTION 3

3.1 $\text{Total parts} = 1 + 2 + 3 = 6$ ✓; $\text{cement} = 0{,}6 \times \frac{1}{6} = 0{,}1\ \text{m}^3$ ✓; $\text{sand} = 0{,}2\ \text{m}^3$ ✓; $\text{stone} = 0{,}3\ \text{m}^3$ ✓ (4)

3.2 Keeping concrete moist (and at a suitable temperature) for several days after it is placed ✓✓, so that it can harden properly, reach its full strength and not crack ✓. (3)

3.3 Any TWO: easy to work with ✓; strong for its weight ✓; renewable; good insulator; attractive appearance (2)

3.4 Any THREE: knots ✓; warping (bowing, cupping, twisting) ✓; shakes / splits ✓; rot; insect damage (3)

3.5 Any TWO: face brick ✓; common (stock) brick ✓; cement (concrete) brick; engineering brick (2)

3.6 It prevents moisture from rising up from the ground into the wall ✓✓ (rising damp). (2)

3.7 Any TWO: paint or varnish ✓; treat with a preservative such as creosote or by pressure treatment ✓; keep dry (2)

3.8 Concrete is strong in compression but weak in tension ✓; steel gives the concrete tensile strength so it does not crack under load ✓. (2)

[20]

QUESTION 4

4.1 Claw hammer: drive and pull out nails ✓; tenon saw: make fine, straight cuts in wood (joints) ✓; brick trowel: spread mortar when laying bricks ✓; cold chisel: cut or chip metal, brick or concrete ✓; builder's line: keep brick courses straight and level ✓ (5)

4.2 Any THREE: check cables and plugs for damage before use ✓; unplug before changing blades or bits ✓; clean after use ✓; store in a dry place; have them serviced; use only for their intended purpose (3)

4.3 Place the level on a flat surface and read the bubble ✓; turn it end for end (180°) in the same position ✓; if the bubble is in the same place both times, the level is accurate ✓. (3)

4.4 Any TWO, each with a use: mortise and tenon ✓, used in door and window frames, tables ✓; dovetail ✓, used in drawers and boxes ✓; butt joint, used for simple frames; halving joint, used for frames (4)

[15]

QUESTION 5

5.1 To spread the load of the building evenly onto the soil ✓ so that the building does not sink or crack ✓. (2)

5.2 $\text{V} = 12 \times 0{,}6 \times 0{,}2$ ✓✓ $= 1{,}44\ \text{m}^3$ ✓ (3)

5.3.1 $\text{Area} = 6 \times 2{,}4$ ✓ $= 14{,}4\ \text{m}^2$ ✓; $14{,}4 \times 55$ ✓ $= 792\ \text{bricks}$ ✓ (4)

5.3.2 $792 \times 1{,}05 = 831{,}6$ ✓ $\rightarrow 832\ \text{bricks}$ ✓ (2)

5.3.3 $832 \times \text{R}4{,}50$ ✓ $= \text{R}3\ 744$ ✓ (2)

5.4 Any FOUR: rafters ✓; purlins / battens ✓; wall plate ✓; ridge ✓; tie beam; king post; fascia board (4)

5.5 To carry the load of the brickwork above a door or window opening ✓✓. (2)

5.6 Any THREE: makes the soil stronger so it can carry the load ✓; prevents settlement and cracks ✓; removes air pockets ✓; reduces water penetration (3)

5.7 Measure 3 m along one line ✓ and 4 m along the other line ✓; adjust until the diagonal between the two points is exactly 5 m, then the angle is 90° ✓. (3)

[25]

QUESTION 6

6.1 (a) visible outlines / edges ✓✓ (b) hidden detail (edges that cannot be seen) ✓✓ (c) centre lines / lines of symmetry ✓✓ (6)

6.2 DPC: damp-proof course ✓; FFL: finished floor level ✓; NGL: natural ground level ✓ (3)

6.3 $4\ 000\ \text{mm} \div 50$ ✓ $= 80\ \text{mm}$ ✓ (2)

6.4 Front view (front elevation) ✓; top view (plan) ✓; side view (side / end elevation) ✓ (3)

6.5 To draw large objects on paper in correct proportion ✓. (1)

[15]

TOTAL: 100

END OF MARKING GUIDELINE